



# A combined experimental–numerical study of the compaction behavior of NaCl

Steffen Brinckmann, Guofeng Gao, Thomas Siegmund\*

School of Mechanical Engineering, Purdue University, United States

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## ABSTRACT

The compaction behavior of NaCl as a model substance is investigated by an integrated experimental and computational approach. The method for characterization of this granular material employs convenient experiments: load–displacement measurements of compaction; measurements of strain on outer circumference of an elastic tubular die; load on bottom and top of the powder compact, as well as compressive strength tests. Related equations for identification of material parameters are derived and are used to characterize powder behavior and powder–die friction. Subsequently, these material parameters are used in simulations with the Drucker–Prager–Cap (DPC) model. For the verification of the computations density distributions are determined based on micro X-ray computer tomography. Good agreement between the spatial density distributions from measurements and simulations is obtained. Restrictions of computer tomography in powder compaction applications are specified. While the study employs NaCl as a model substance, the approach is applicable to a wider array of granular substances.

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## 1. Introduction

This study described an integrated approach for numerical simulations of powder compaction and the detailed experimental verification of model predictions through density measurements were based on high resolution  $\mu$ CT scans. The study is conducted with the goal to aid the development of advanced processing models for powder-based materials.

A numerical simulation model is the centerpiece of the approach to the manufacturing process analysis. In particular, the Drucker–Prager–Cap (DPC) model is employed. This model, originally developed for geotechnical applications, Sandler et al. [1], is a frequently used continuum model in powder compaction, Aydin et al. [2], Falgon et al. [3], Michrafy et al. [4], Cunningham et al. [5], Wu et al. [6,7] and Frenning [8]. Solutions to boundary value problems related to compaction processes are computed with the finite element method. A numerical model alone, however, is not useful. In conjunction with such a model a process of material parameter identification is needed.

Two approaches to the determination of material parameters are possible. In the inverse modeling approach material parameters are identified from best fit between the target experiments and numerical simulation. This approach, however, often leads only to agreement between experiments and simulations in the specific experimental set-up used for material identification. The second and more general approach to material parameter identification employs simple experi-

ments for parameter identification. The results of these experiments in conjunction with analytical expressions are used to identify the material parameters. This approach is employed in the present study.

Two approaches proposed for the determination of the material parameters appear as relevant, Aydin et al. [2] and Es-Saheb [9]. Aydin et al. [2] assumed that the slope of the yield surface in stress space during loading is identical to the internal friction angle  $\beta$  for all cylindrical dies and neglected the effect of powder–die friction during loading. In the present study we will use a more general approach. It includes the determination of the coefficient of friction and does not make any assumptions about the slope of the yield surface in stress space. The equations employed can be used to characterize granular materials under irreversible deformation directly without the need of inversely fitting simulation results to experimental data. Es-Saheb [9] used a pin-type transducer to measure the radial stress at the powder–die interface. This approach results in a difficult experimental set-up and disturbance of the die-wall friction due to the pin. In the present study measurements of the radial stress are obtained from the elastic expansion of an elastic tubular die. The deformation of the elastic die is measured on a continuous basis by the use of a strain gauge attached to the die, and the die can be reused. The use of a tubular die was also proposed in Nielsen et al. [10]. However, these authors used the geometric measures of the permanent (plastic) deformed tube shape to infer the pressure in the powder. Both approaches provide alternatives to the use of a triaxial test cells for powder compaction, e.g. Schneider and Cocks [11].

Simulations of the cylindrical compaction experiment are conducted. Subsequently, the geometry of the pressing set-up is modified to a complex geometry for which no theoretical solution exists. Any

\* Corresponding author. 585 Purdue Mall, West Lafayette, IN 47907. Tel.: +1 765 494 9766; fax: +1 765 496 7536.

E-mail address: [siegmund@ecn.purdue.edu](mailto:siegmund@ecn.purdue.edu) (T. Siegmund).

numerical model requires validation to establish its range of validity and confidence in model predictions. Validations of model predictions are conducted on two levels. First load–displacement results from experiments and simulations are compared. Subsequently, the density distribution in powder compacts is assessed through X-ray microscale tomography ( $\mu$ CT) scans of powder compacts.

The use of X-ray systems for density measurements has been of considerable interest in powder technology. Early studies employed low resolution systems. Compaction processes were assessed by either tracing motion of periodically placed tracer particles, Aydin et al. [2], or by correlating the X-ray absorption spectra to the density, Phillips and Lannutti [12], Bonnenfant et al. [13], and Kong and Lannutti [14]. More recently, high resolution systems have become of interest as well. These systems are of interest due to their high spatial resolution allowing for the potential to obtain images with not only average density distributions but to trace individual small particles. Sinka et al. [15] employed X-ray computed tomography (CT) to study pharmaceutical powders and obtained densities from comparing spatial absorption distribution to that of compacts of calibrated density. Fu et al. [16] use such a system to study powder compaction with an in-situ compaction cell in a  $\mu$ CT system. This study again employed tracer particles. In Wu et al. [7] micro-tomography was employed to qualitatively describe density distribution and crack formation associated with capping of pharmaceutical tablets but no quantitative data for the density distribution was obtained. In the present study, 3D  $\mu$ CT scans were processed by image analysis tools, such that first the void space map was determined, and based on the void distribution the density distribution is determined. Imaging of compacted NaCl via 3D  $\mu$ CT scans was also used by Marmottant et al. [17] in order to establish microstructure correlations.

The present study employs NaCl as a model material. This material is often employed to test pharmaceutical production processes (e.g. Hardman and Lilley [18], Lordi et al. [19], Sonnergard [20], Simon and Guigon [21], Picker [22], Alderborn [23], Doelker and Massuelle [24]). These past studies have investigated some of the basic mechanical properties of NaCl compacts but a complete characterization of NaCl compaction through the Drucker–Prager–Cap model has not been conducted.

## 2. Materials and methods

### 2.1. Model material

NaCl is a crystalline material with a cubic structure and possesses low shear strength on the (110) and (100) planes, Davidge [25]. Fig. 1(a) shows the normalized histogram of the grain size distribution. The bulk and the theoretical density are  $\rho_0 = 1154 \text{ kg/m}^3$  and  $\rho_d = 2165 \text{ kg/m}^3$ , respectively, [26]. Fig. 1(b) shows optical micrographs of individual NaCl grains. Typical compacts of NaCl are depicted in Fig. 2(a), and Fig. 2(b) shows compressed NaCl at high density. Individual grains are closely stacked and many grains contain cracks. While NaCl grains initially are cubic, Fig. 1(b), nearly all grains have lost their initial shape due to shearing after compression.

### 2.2. The Drucker–Prager–Cap model

The elastic deformation of the powder compact is characterized as linear and described by Hook's law. The plastic regime is defined by two yield surfaces, Fig. 3, in terms of hydrostatic pressure  $p = 1/3 \text{ trace}(\boldsymbol{\sigma})$  and the von Mises stress  $q = \sqrt{3/2}(\mathbf{S} : \mathbf{S})$ ,  $\mathbf{S} = \boldsymbol{\sigma} - p\mathbf{I}$ . The shear yield surface captures sliding of grains with respect to each other. It is defined by a line in  $p-q$  space with the cohesion  $d$  as the offset from the origin and a slope corresponding to the angle of internal friction ( $\tan \beta$ ):

$$q = d - p \tan \beta. \quad (1)$$

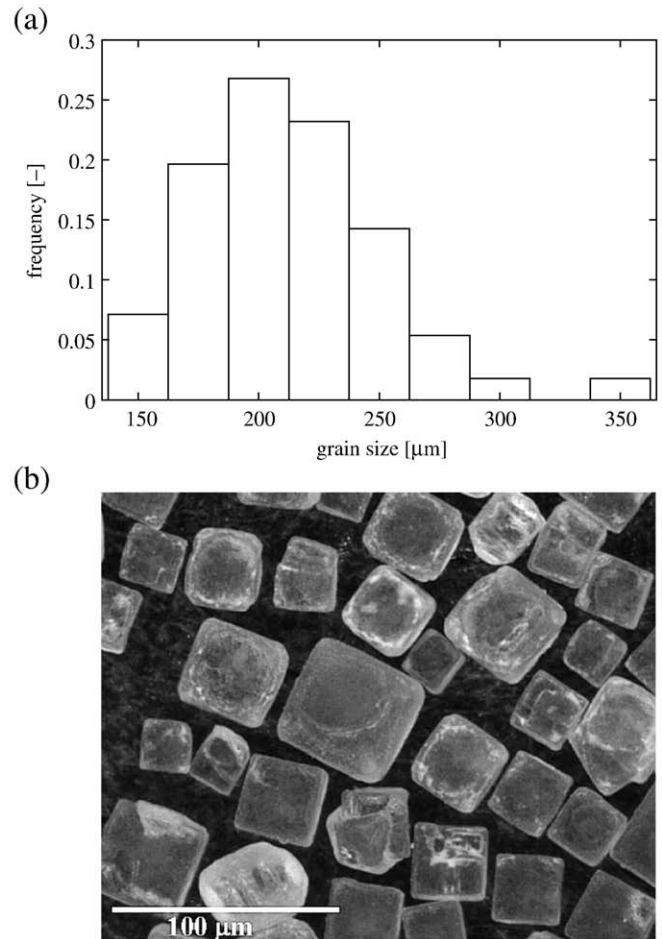


Fig. 1. (a) Normalized histogram of the grain size distribution in the NaCl powder considered, (b) micrograph of sample NaCl grains.

Cohesion is dependent on the relative increase in density,  $d = d(\rho_{\text{rel}})$ . The relative increase density  $\rho_{\text{rel}}$  is defined as the ratio between the current density  $\rho$  and the initial bulk density  $\rho_0$ ,  $\rho_{\text{rel}} = \rho / \rho_0$ . The maximum relative increase in density over bulk density is  $\rho_{\text{rel}}^* = \rho_d / \rho_0 = 1.874$ .

The cap yield surface characterizes irreversible deformation due to compaction. The cap is defined by two characteristic pressures  $p_a$  and  $p_b$ . Equivalently, it is defined by the use of the cap eccentricity factor  $R$ :

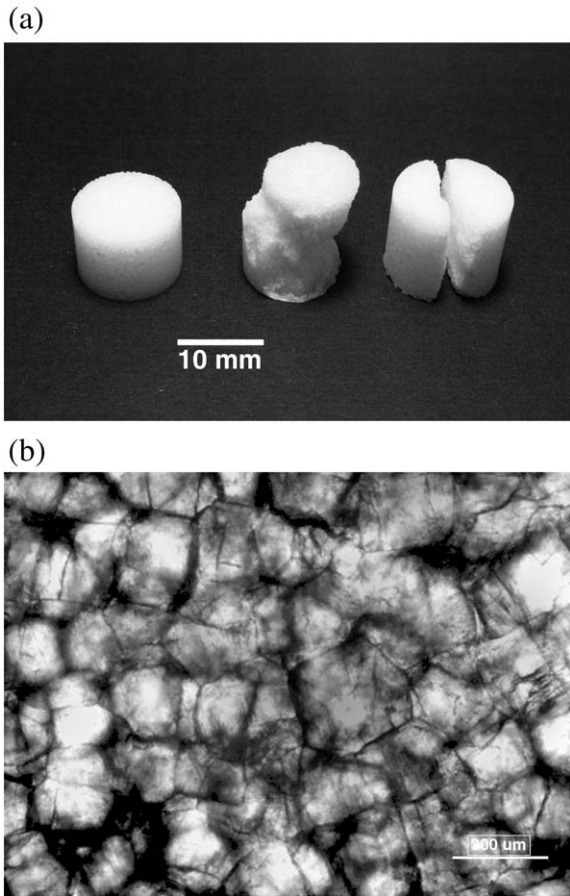
$$p_a - p_b = R(d - p_a \tan \beta) \quad (2)$$

where  $(d - p_a \tan \beta)$  is the shear strength at  $p_a$ . Finally, a hardening connects the evolution of the characteristic pressure  $p_b$  to the plastic volumetric strain  $\varepsilon_{\text{vo}}^{\text{pl}}$ .

### 2.3. Compaction experiments

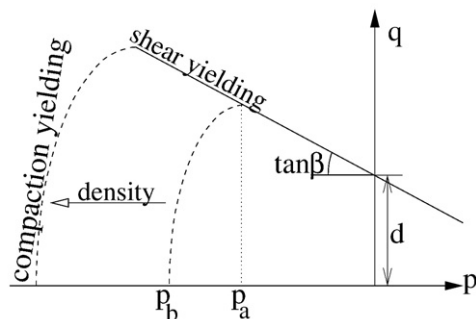
Powder compaction experiments were conducted in single action press configurations and by the use of cylindrical dies. A manual press (Carver Inc.) with a thick walled die was used to obtain compacts for a large range of compaction stress. A MTS 858 Mini Bionix servohydraulic test system was used for instrumented compaction experiments with both a thick and a thin-walled die. These compaction experiments employed a loading rate of 7.5 kN/min.

The thick walled die, Fig. 4(a), possesses an inner radius of  $r_i = 6.5 \text{ mm}$  and an outer radius of  $r_o = 25.3 \text{ mm}$  while the thin-walled die, Fig. 4(b), possesses an inner radius of  $r_i = 6.5 \text{ mm}$  and an outer radius of  $r_o = 9.5 \text{ mm}$ . Three different punch geometries were considered,

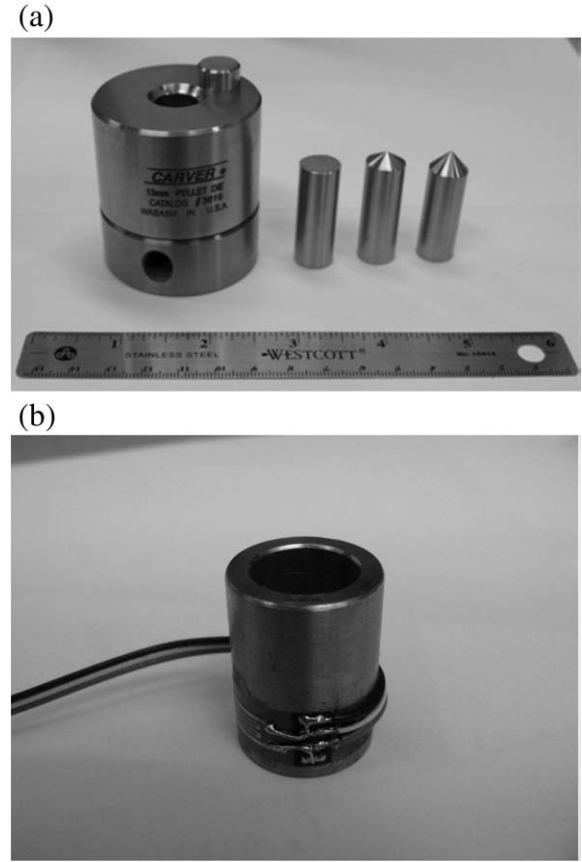


**Fig. 2.** (a) A typical compact, a uniaxial compression specimen and a Brazilian disk test specimen, (b) optical image of compressed NaCl (compaction stress 830 MPa).

**Fig. 4(a).** First, experiments were conducted using a flat punch. Subsequently, punches with 30° and 45° angles were used, too. All three punches possess nearly the same weight. Dies are made of steel ( $E^s = 200$  GPa,  $\nu^s = 0.33$ ). A steel disk (6.7 mm height) was used as an insert on the bottom of the dies. **Fig. 5(a)** defines the details of the geometric features relevant for the compaction process. The dimensions  $h_0$  and  $h$  are die fill height and current compact height, respectively. The  $z$ - and  $r$ -axes define axial and radial positions, respectively. The  $z$ -axis denotes the distance from the bottom of the powder (or compact). Dies were filled with 2.5 g of NaCl. The die fill height was determined as  $h_0 = 14.97$  mm for the case of the flat punch. This initial compaction is due to the weight of the punch. The actual compaction process starts at a density higher than that of the bulk material,  $\rho_{rel,0} = 1.135$ . Compact densities were determined as ratio between mass and volume after ejection from the die.

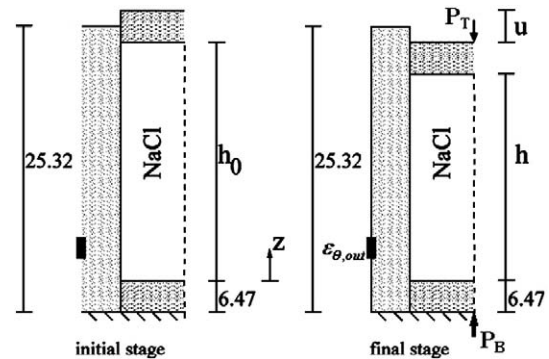


**Fig. 3.** Drucker-Prager-Cap model in stress space, (hydrostatic pressure  $p$ , van Mises stress  $q$ ).



**Fig. 4.** Photographs of (a) thick walled die and punches; (b) elastic tubular die with strain gauge.

The fundamental compaction response is determined from measurements the punch  $u$  (as measured by the displacement transducer of the test system) and the total load applied  $P_T$  (measured by a load cell), **Fig. 5(b)**. The compaction stress is then  $\sigma_T = P_T / (r_i^2 \pi)$ . The load transmitted through the compact was measured by employing a second load cell placed such that only the force  $P_B$  from the die insert at the bottom was recorded, **Fig. 5(b)**. Then, the relationship between the stress transmitted to the bottom of the die  $\sigma_B = P_B / (r_i^2 \pi)$  and the compaction stress  $\sigma_T$  is obtained. The relationship between the compaction stress  $\sigma_T$  and the radial stress  $\sigma_r$  at the compact-die-wall interface can be determined based on measurements using the elastic tubular die, **Fig. 5(b)**. To this purpose a 120  $\Omega$  strain gage was attached on the outside of the elastic tubular die, **Fig. 4(b)**, such that circumferential strains on the die surface  $\epsilon_{\theta,out}$  are recorded. The midsection



**Fig. 5.** Geometry of the compaction process: (a) initial stage, (b) final stage. The relevant measurement data are indicated in (b): displacement  $u$ , force on top of die  $F_T$  force at bottom of die  $F_B$ , and circumferential strain  $\epsilon_{\theta,out}$  at die outside.



of the strain gage was located at a distance of 4.16 mm from the bottom of the powder.

#### 2.4. The shear yield surface

Uniaxial compression tests and Brazilian disc tests were conducted on cylindrical compacts of a wide range of densities. These measurements were conducted using a MTS 858 Mini Bionix servohydraulic test system under displacement control with a displacement rate of 0.02 mm/s. Fig. 2(a) depicts typical specimen post-test.

Samples for the Brazilian test do not place any requirement on the height  $h$  to diameter  $d$  ratio, and  $h/d = 1$  was chosen. Samples for uniaxial compression testing, however, require  $h/d > 2$  such that shearing along the  $45^\circ$  angle can occur. Such high aspect ratio compacts were obtained by combined pressing of two pre-compacted specimens. No interfaces of relevance developed in such specimens since the pre-compaction stress was low. From the initial slope of the load–displacement record of the uniaxial compression test the Young's modulus was determined. From the maximum loads of the uniaxial compression and Brazilian disk test ( $F_{U,uni}$  and  $F_{U,braz}$ ), respectively, corresponding values of  $(p, q)_{uni}$  and  $(p, q)_{braz}$  are obtained, Kraft et al. [27]:

$$(p, q)_{uni} = \left( \frac{1}{3} \frac{F_{U,uni}}{\pi r_i^2}, \frac{F_{U,uni}}{\pi r_i^2} \right), \quad (3)$$

$$(p, q)_{braz} = \left( \frac{2}{3} \frac{F_{U,braz}}{\pi b r_i}, \sqrt{13} \frac{F_{U,braz}}{\pi b r_i} \right). \quad (4)$$

The lines through the points  $(p, q)_{uni}$  –  $(p, q)_{braz}$  characterize the shear yield surface.

#### 2.5. Die-wall coefficient of friction and hardening law

The data sets  $(\sigma_r - u, \sigma_r - \sigma_B, \sigma_r - \varepsilon_{\theta,out})$  are needed to obtain the coefficient of friction  $\mu$  between die and powder compact, and the hardening law. Following the Janssen–Walker theory (the assumption of plane cross sections) the axial stress  $\sigma_z(z)$  and the coefficient of friction  $\mu$  for a Coulomb friction law ( $\tau = \mu \sigma_r$ ) are [6]:

$$\sigma_z = \sigma_B \left( \frac{\sigma_r}{\sigma_B} \right)^{\frac{2}{h}} \quad (5)$$

$$\mu = \frac{r_i}{2h} \frac{\sigma_B}{\sigma_r} \left( \frac{\sigma_r}{\sigma_B} \right)^{\frac{2}{h}} \ln \frac{\sigma_r}{\sigma_B} \quad (6)$$

with  $\sigma_r(z)$  the radial stress. The stress transmission ratio  $\kappa$  is [6]:

$$\kappa = \frac{\sigma_r}{\sigma_z} = \frac{r_i}{2h\mu} \ln \left( \frac{\sigma_r}{\sigma_B} \right) = \frac{\sigma_r}{\sigma_B \left( \frac{\sigma_r}{\sigma_B} \right)^{\frac{2}{h}}}. \quad (7)$$

For the elastic tubular die, the radial stress  $\sigma_r$  at the compact–die interface is determined from the measured circumferential strain  $\varepsilon_{\theta, out}$ . Following the analysis for an elastic thick walled pressure vessel with inner radius  $r_i$  and outer radius  $r_o$ ,  $\sigma_r$  relates to the circumferential stress at the outside die surface  $\sigma_{\theta,out}$ :

$$\sigma_r = \frac{r_o^2 - r_i^2}{2r_i^2} \sigma_{\theta,out}. \quad (8)$$

Hooke's law relates  $\varepsilon_{\theta,out}$  to  $\sigma_{\theta,out}$ :

$$\sigma_{\theta,out} = E^s \varepsilon_{\theta,out} + v^s \frac{F(z)}{\pi(r_o^2 - r_i^2)}. \quad (9)$$

$F(z)$  is the normal force in the die resulting from friction between compact and die. The function  $F(z)$  follows from integration of the shear stresses from the top of the powder compact  $z = h$  to location  $z$ :

$$\begin{aligned} F(z) &= 2\pi r_i \int_z^h \tau_z dz^* = 2\pi r_i \mu \kappa \sigma_B \int_z^h \left( \frac{\sigma_r}{\sigma_B} \right)^{\frac{2}{h}} dz^* \\ &= \pi r_i^2 \left[ \sigma_B \left( \frac{\sigma_r}{\sigma_B} \right)^{\frac{2}{h}} - \sigma_r \right]. \end{aligned} \quad (10)$$

Consequently, an average axial stress  $\bar{\sigma}_z$  and an average hydrostatic pressure  $\bar{p}$  in the compact are:

$$\bar{\sigma}_z = - \frac{\sigma_B - \sigma_r}{\ln \left( \frac{\sigma_r}{\sigma_B} \right)} \quad (11)$$

$$\bar{p} = \frac{1}{3} (\sigma_z + 2\kappa \sigma_r). \quad (12)$$

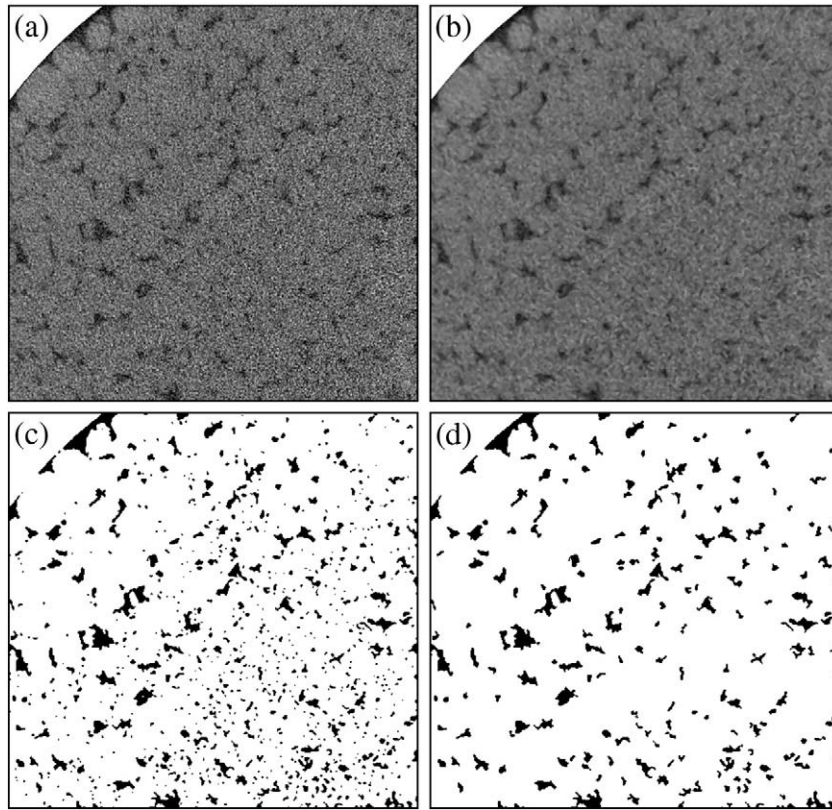
The permanent volumetric strain is:

$$\varepsilon_{vol}^p = \frac{1}{h_0} \left( u - \frac{\sigma_r}{C} \right). \quad (13)$$

In Eq. (13) the radial and circumferential strain components are neglected since these are orders of magnitude smaller than the axial strain. Thus, the hardening law, the relationship between  $\bar{p}$  and  $\varepsilon_{vol}^p$ , is defined. The elastic unloading stiffness  $C$  can be determined from the unloading curve of the load–displacement data. For the present study, a value of  $C = 561.6$  MPa/m was employed which represents compacts at a medium high compaction stress. The choice of the value of  $C$  little influences results since elastic deformation is small.

#### 2.6. Measurement of density distribution through $\mu$ CT

Compacts were placed into a computed tomography system “Scan-coMedical  $\mu$ CT 40” which provides a beam energy of 70 kVp and flux of 145  $\mu$ A. The spatial resolution of the system is 6  $\mu$ m. Compacts were placed into the sample holder without any fixative. To construct a three-dimensional image, the structure is divided into planes or slices. Each plane is scanned from every direction between  $0^\circ$  and  $180^\circ$ . Each of these directional scans leads to a projection of the integrated X-ray absorption along the line perpendicular to the scan direction. These absorption integrals are then used to compute the absorption map of the slice by correlating results from all directional scans. Once one slice is finished the next slice is started. The retrieved images have a size of  $2048 \times 2048$  pixel per slice scanned and are transported in jpeg format. The image processing steps undertaken are exemplified for the case of a compact made with a flat punch and a relative increase in density of  $\rho_{rel} = 1.5$ . A typical image slice of this compact is shown in Fig. 6(a). In this image, the contrast between the void space and the dense solid is low, especially if the relative density is high. Image processing is used to improve the measurement. The images obtained from  $\mu$ CT are processed using the image package of Matlab. First, the two-dimensional median filter is applied, Fig. 6(b), to reduce the noise in the grayscale images. Next, the image is converted from grayscale to black–white with a threshold of 32%, Fig. 6(c), such that black pixels accurately represent the void space. The determination of the threshold value was made using scans of low relative density where individual grains could accurately be distinguished. Finally, white and black noise of 15 and 20 pixel size or smaller are deleted, respectively, such that an image depicting the void space is obtained, Fig. 6(d). Following image processing, the spatial relative density distribution is determined. The void volume fraction  $f_v$  is the ratio between the number of black pixels to the total number of pixels in a given domain, and  $\rho_{rel}$  is obtained as  $(1 - f_v)(\rho_d / \rho_0)$ . Assuming axisymmetric



**Fig. 6.** Image processing steps: (a) original image from mCT scan, (b) after application of median filter, (c) after grayscale to black/white conversion, (d) after removal of white and black noise. The image (d) represents the final image.

conditions the density distribution is then given in dependence of location in the  $r$ - $z$  plane. Finally, an overall relative density can be computed by averaging over the entire compact volume. For the calibration of the image processing procedure, the average density obtained from image processing is scaled such that it matches the relative density determined based on mass and volume of the individual compact investigated. This procedure was chosen as the goal of the present investigation was the determination of the spatial density distribution.

### 2.7. Numerical simulations

The numerical simulations are executed using the Drucker–Prager–Cap model of the ABAQUS/Standard finite element program. This implementation of the DCP model uses a transient curve between the shear and compression yield surfaces. This transition surface is characterized by a coefficient  $\alpha$ , with  $\alpha=0$  implying no transient surface, and  $\alpha=1$  eliminating the cap surface completely. All present simulations employ  $\alpha=0.03$ . The implementation of the DPC model in ABAQUS also includes a non-associated flow surface compared to the yield surface which leads to a non-symmetric stiffness matrix.

The finite element model employs 4 node axisymmetric elements (CAX4 in ABAQUS). For the case of the flat punch, the elements were of rectangular shape. For the cases with cone shaped punches the elements were of trapezoidal shape. In all cases the initial volume of the elements representing NaCl was constant, and a spatially uniform initial density distribution with  $\rho_{\text{rel},0}=1.135$  was used. This initial condition ignores that for the cone punches an initially inhomogeneous density distribution might exist. Since the density changes during compaction are much larger than this initial inhomogeneity it can be argued that the approach taken will lead to acceptable results, as shown from subsequent comparison of experimental and numerical results. Three contact surfaces are present in the model. The

interaction between powder and punch as well as between powder and bottom die are assumed as frictionless. The interaction between powder and the vertical die wall is assigned a constant coefficient of friction. The thick walled die is modeled as a rigid body, while the thin-walled die is considered as an elastic structure.

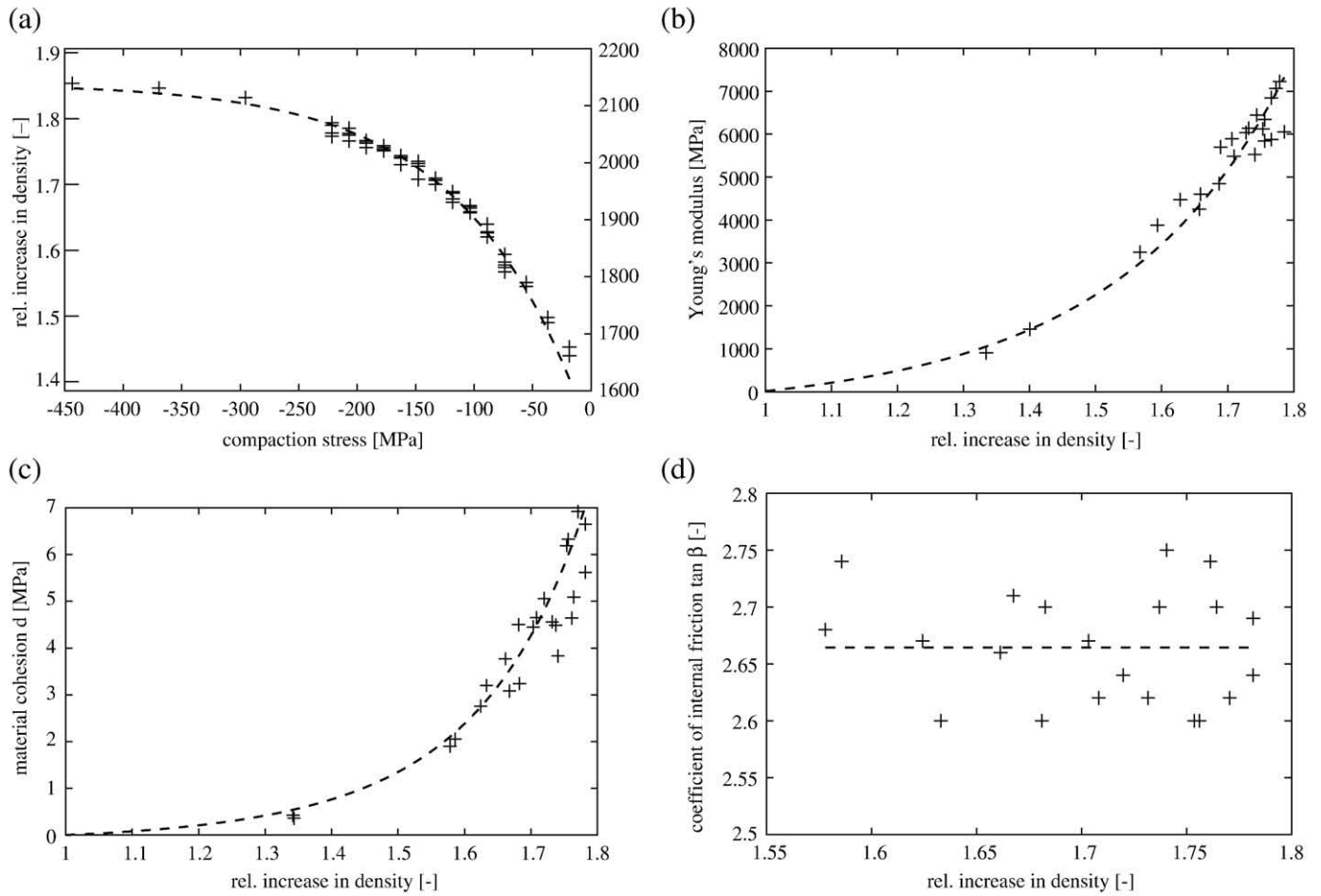
The remaining material properties are as follows. A Poisson's ratio of 0.12 is used. The initial cap yield surface position is 0.0, and the flow stress ratio 1.0. The geometry factor  $R$  is assumed as 0.6 based on similar approaches in Refs. [2,3,6,8].

The relative density was calculated from computed strain values as  $\rho_{\text{rel}} = \rho_{\text{rel},0} / \exp(\epsilon_{\text{vol}}^p)$ . Material parameters are defined as dependent on the relative density. In the finite element code this is accomplished through a user variable in the subroutine USDFLD. To post-process computations, relative densities were also computed in a user subroutine UVARM in conjunction with a user defined output variable.

## 3. Results and discussion

### 3.1. Experimental results

The dependence of the density of NaCl compacts on compaction stress  $\sigma_T$  is shown in Fig. 7(a). An exponential fit  $\rho_{\text{rel}} = \rho / \rho_0 = 1.85 - 0.64 \exp(-0.0097\sigma_T)$  with a root mean square value of  $R=0.012$  characterizes the experimental data. No experimental data points are available for very lowest compaction stresses ( $\sigma_T < 20$  MPa) since such powder compacts could not be handled due to a lack of strength at low compaction levels. Under high compaction stress the fully dense NaCl is approached. The Young's modulus was determined as the linear elastic slope of stress–strain records of compression tests in the longitudinal direction of freestanding powder compacts. The dependence of Young's modulus on  $\rho_{\text{rel}}$  is shown in Fig. 7(b). The fit function  $E = -1049 + 689.2 \cosh(\rho_{\text{rel}}^2)$  [MPa],  $R=487.8$  MPa, describes the experimental data. The dependence of the material cohesion  $d$  on  $\rho_{\text{rel}}$



**Fig. 7.** (a) Relative increase in density and absolute density in dependence on compaction stress; (b) Young's modulus in dependence of relative increase in density, (c) material cohesion in dependence of the relative increase in density, and (d) internal friction angle in dependence of the relative increase in density.

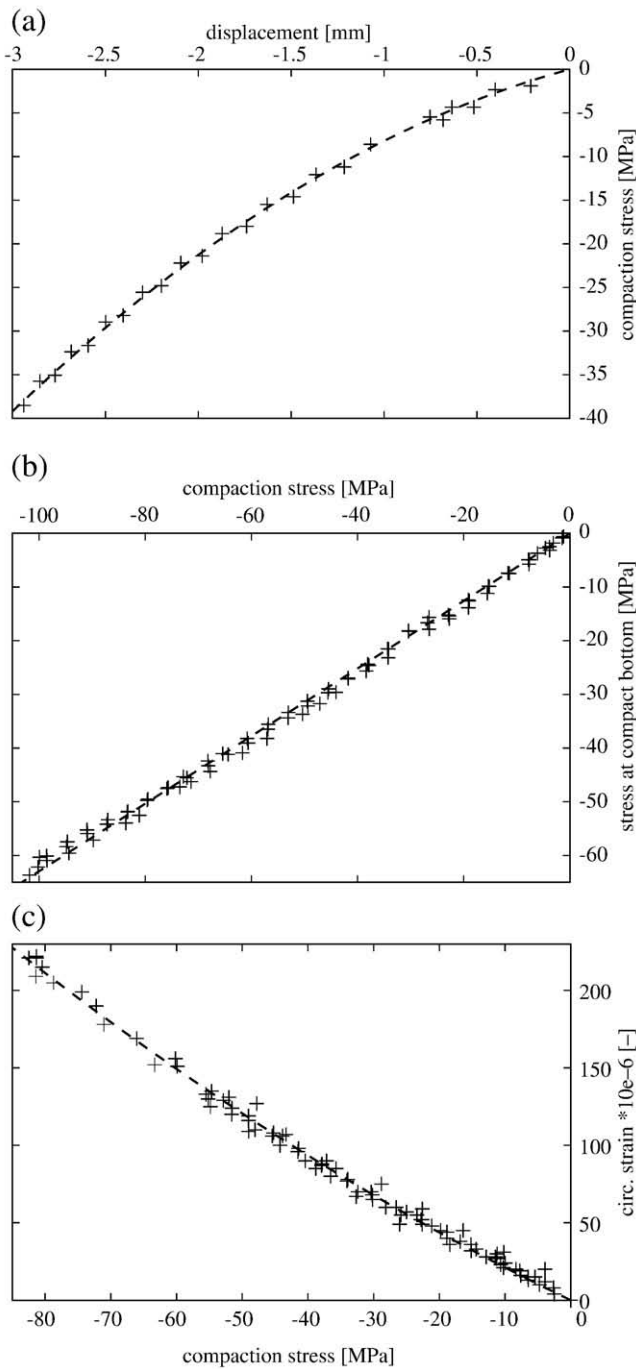
is given in Fig. 8(c). Experimental data can be fitted as  $d = 0.21 [\cosh(\rho_{\text{rel}}^{2.5}) - \cosh(1.0)]$  [MPa],  $R = 0.663$  MPa. The coefficient of internal friction was found to be not depend on density,  $\tan \beta = 2.66$ ,  $R = 0.047$ , see Fig. 7(d).

The relationship between compaction stress  $\sigma_T$  and the punch displacement  $u$ , Fig. 8(a), during powder compression was recorded following the set-up shown in Fig. 5. A fit to the experimental data is  $\sigma_T = -2.43u^2 + 5.79u$  [MPa],  $R = 0.53$  MPa. Using the set-up with two load cells the relationship between applied and transmitted load is determined, Fig. 8(b). Data is characterized as  $\sigma_B = 0.63\sigma_T$  [MPa],  $R = 1.07$  MPa. Finally, the relationship between compaction stress and circumferential strain is shown in Fig. 8(c). The experimental data are well described by the fit  $\varepsilon_{\theta, \text{out}} = 10^{-6}(0.008\sigma_T^2 - 2.034\sigma_T)$ ,  $R = 4.494$ . From Eq. (6), the coefficient of friction for NaCl in combination with the die employed in the study follows, Fig. 9(a). The coefficient of friction was determined as  $\mu = 0.34$  at the onset of compaction and increases monotonically with compaction to  $\mu = 0.52$ . The average coefficient of friction is  $\bar{\mu} = 0.43$ , a value subsequently considered in the numerical simulation. If the normal force in the die is neglected in the calculation of the coefficient of friction, overall slightly higher values of the coefficient of friction are obtained. The stress transmission ratio  $\kappa$  is shown in Fig. 9(b). This parameter varied only little during compaction. Finally, the hardening curve ( $\varepsilon_{\text{vol}}^p - p$ ) is determined, Fig. 9(c). The experimental data are well described by the fit function  $p(\varepsilon_{\text{vol}}^p) = -12.66 [\exp(-4.82 \varepsilon_{\text{vol}}^p) - 1.0]$  [MPa],  $R = 1.19$  MPa. The hardening law, i.e. relationship between hydrostatic pressure and volumetric strain, is a central piece of numerical studies of powder compaction. Wu et al. [7] proposed the use of an expo-

nential a law of type  $p(\varepsilon_{\text{vol}}^p) = A \exp(B \varepsilon_{\text{vol}}^p)$ . Using this approach, the present experimental data can be approximated by  $p(\varepsilon_{\text{vol}}^p) = -7.305 \exp(-5.688 \varepsilon_{\text{vol}}^p)$  [MPa],  $R = 4.76$  MPa. With the exponential hardening law the root mean square error is larger mostly due to the fact that the exponential law of Wu et al. [7] introduces a non-zero threshold hydrostatic pressure at zero volumetric strain. Differences between these two formulations appear, however, as minor as long as the volumetric strain is below a value of 0.3.

### 3.2. Numerical simulations – global response

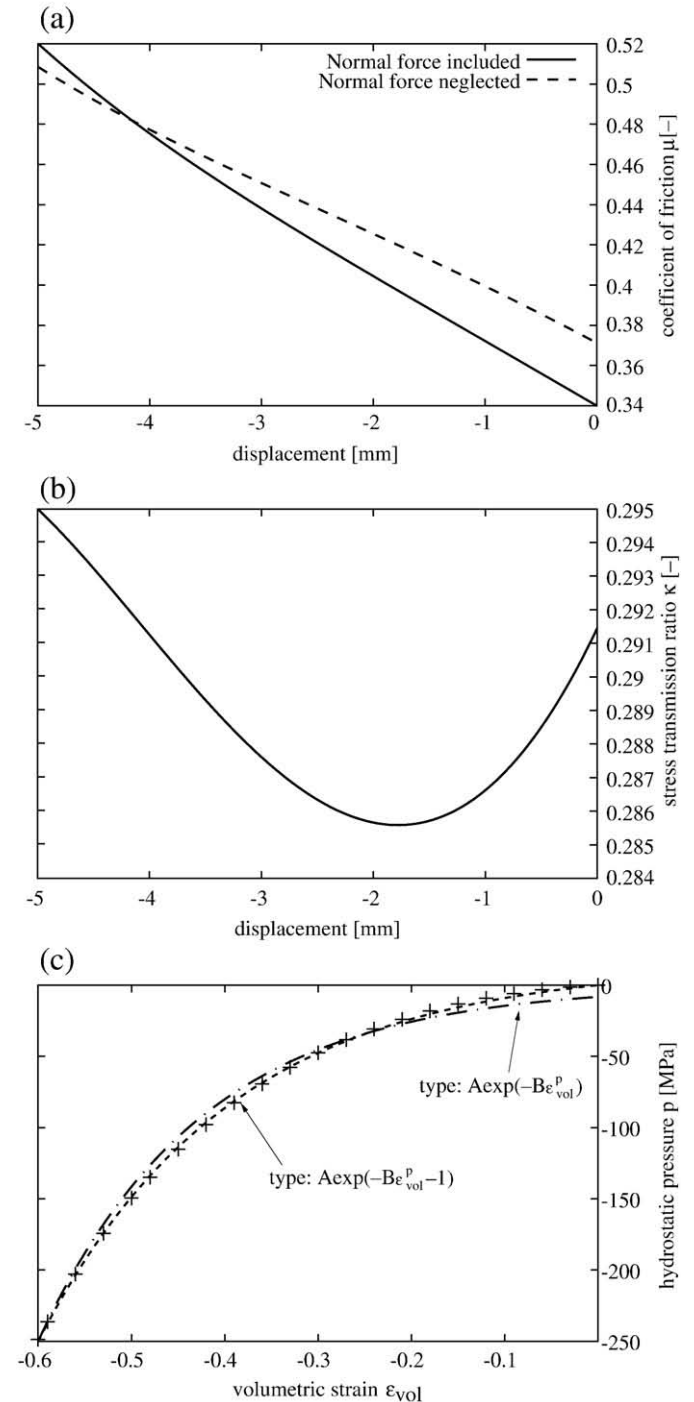
Initially, we consider the experimental and numerical results of the loading-unloading response for the case of compaction with a flat punch. The loading and unloading curves were already employed to determine the material parameters of the DPC model. However, in that derivation simplifying assumptions were made in regard to the complete stress state expected, i.e. plane cross sections remain planar. Finite element simulations do not make this assumption, and therefore, a comparison of the experimental loading-unloading data to the simulation results is a suitable test for the suitability of the material parameters. Fig. 10(a) depicts the comparison between the numerical and experimental results for a flat punch. The initial parts of the loading curves show good agreement. However, during the later part of loading, the load magnitude in the experimental data increases discontinuously, an event not captured by the simulation. This event appears not to be an artifact as it has been observed under many conditions in the experiments. Therefore, the 'jump' is a characteristic property of the compression of NaCl. Two events can potentially



**Fig. 8.** (a) Compaction stress–displacement curve for compression of powder in the thick walled die (data from 2 individual experiments), (b) stress transmitted to the compact bottom in dependence of the compaction stress (data from 3 individual experiments), (c) circumferential strain in dependence of compaction stress (data from 6 individual experiments).

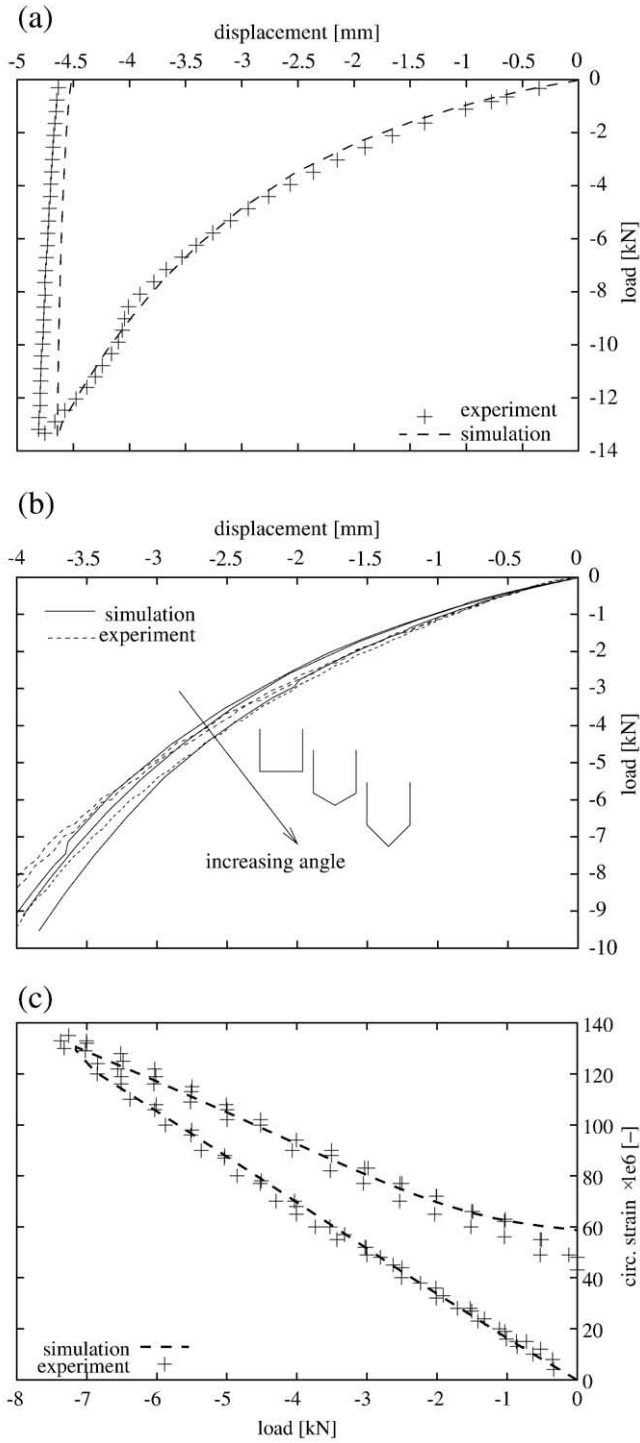
contribute to this instability. During compaction grains enter into contact as load is applied, connect and develop force chains which primarily transfer the load. As the load is increased the force chains can buckle and load is redistributed. Due to buckling the force chain loses its load bearing characteristics and other chains develop due to rearrangement of particles. Gethin et al. [28] have employed a deformable element approach with interface elements to study the behavior of a few grains numerically. That investigation demonstrates the mechanics of such instabilities. A second possible mechanism for instabilities is due to grain shearing. In particular, NaCl grains

shear along the 100 or 110 direction, Hardman and Lilley [18], under relative low stresses. Both processes can contribute to discontinuities in the compaction response, however, for the present experiments it is estimated that grain fracture is significant, see the experimental evidence in Fig. 3. However, a continuum model is not able to capture either effect, i.e. rearrangement or fracture of discrete grains. Due to deviation in the later part of the load response, the unloading curves obtained numerically and experimentally are shifted relative to each other. However, the slopes of the unloading curves agree. Comparing the loading curves for different punch angles, Fig. 10(b), it is found that both experiments and simulations show an increase in the compression load with an increasing angle of the punch. For



**Fig. 9.** (a) Coefficient of friction, (b) stress transmission ratio, and (c) hardening law.





**Fig. 10.** Comparison of prediction (simulation) and experimental data: (a) loading-unloading curve for the case of a flat punch, (b) loading curves for punches with angles 0°, 30° and 45°, (c) circumferential strain in dependence of applied load for the case of a flat punch.

a displacement of 3.0 mm the experiments lead to loads 4.93 kN, 4.99 kN, and 5.42 kN for the 0°, 30° and 45° punch angles, respectively. The simulations predict loads as 4.82 kN, 5.08 kN, and 5.61 kN for the 0°, 30° and 45° punch angles, respectively. Therefore, both simulations and experiments show that the increase in compaction load between 0° and 30° is significantly smaller than the increase between 30° and 45°. Fig. 10(c) shows results of the circumferential strain at the strain gage location obtained from computation on the thin-walled die, and compares these data to the experimentally determined strain gage

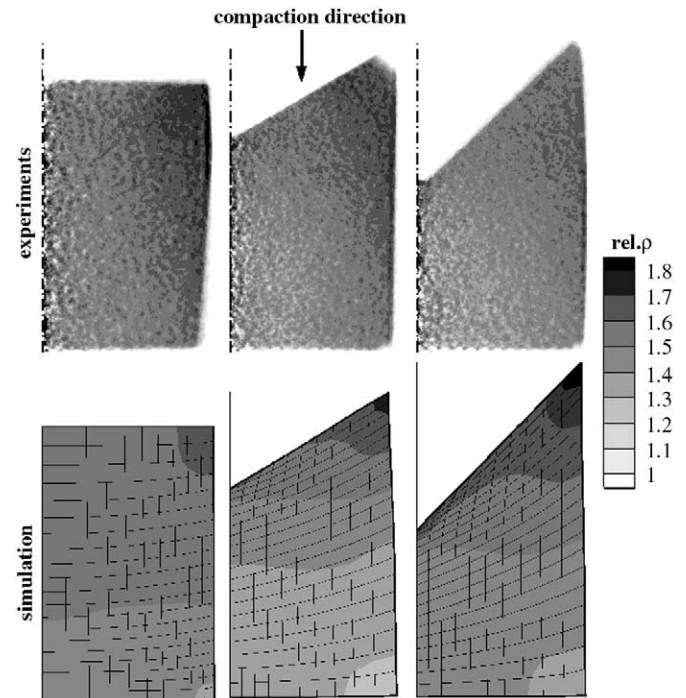
data. During loading, the circumferential strain increases linearly with load. Initially, upon load reversal the circumferential strain decreases linearly but with a smaller slope than during loading. Consequently, a residual circumferential strain  $\epsilon_{\theta, \text{out}}^{\text{res}}$  remains after complete unloading. In particular, for the simulation close to full unloading the circumferential strain decreases nonlinearly and the dependence of the circumferential strain on the load is further reduced. The measured residual circumferential strain ( $\epsilon_{\theta, \text{out}}^{\text{res}} = 50 \times 10^{-6}$ ) is somewhat higher than the predicted circumferential residual strain ( $\epsilon_{\theta, \text{out}}^{\text{res}} = 58 \times 10^{-6}$ ) at the strain gage location. Following Eqs. (8) and (9), the residual radial stress  $\sigma_r^{\text{res}}$  in the compact at the compact-die interface can be obtained as ( $F(z) = 0$ ):

$$\sigma_r^{\text{res}} = \frac{r_o^2 - r_i^2}{2r_i^2} E^s \epsilon_{\theta, \text{out}}^{\text{res}} \quad (14)$$

The experimental data lead to  $\sigma_r^{\text{res}} = 5.7$  MPa, while the simulation predicts  $\sigma_r^{\text{res}} = 6.6$  MPa. Overall, the predictions of the simulations are verified by the experiment on a global level, i.e. the macroscopic characteristics agree.

### 3.3. Numerical simulations – local response

Fig. 11 compares the spatial distribution of the relative increase in density due to compaction as determined from  $\mu$ CT and image processing to that from the finite element analysis including the simulation of the ejection process. The compaction experiments analyzed considered a punch displacement of 4.0 mm. For the flat punch an average relative increase in density of  $\rho_{\text{rel}} = 1.5$  was obtained. The highest density is found at the rim of the top of the compact while the lowest density is at the rim of the bottom of the compact. The differences in the relative increase in densities at these two locations relative to the average density and to that in the center of the compact are well predicted. As the punches with the cone shapes are considered, the deviation from the average density increases with the increase in cone angle. This behavior is again found both in experiments and simulations. A more detailed comparison between measured density



**Fig. 11.** Comparison of density distribution between experiments evaluated by CT imaging and FE predictions. Punches with 0°, 30° and 45° angles.



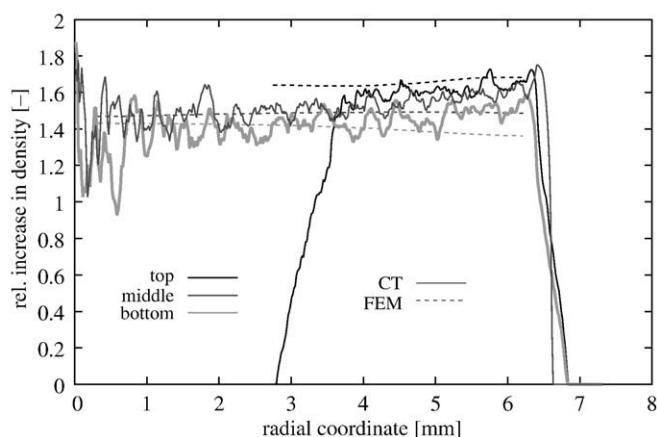


Fig. 12. Predicted and measured radial dependence of relative increase in density distributions for the 45° punch.

distribution and the numerical predictions becomes possible if profiles of relative increase in density are considered for constant height locations in the compact. Fig. 12 depicts profiles of relative increase in density at 4%, 41% and 77% of the compact height for the case of the 45° punch. In this figure, experimental data show a slight increase in density with increase in the radial coordinate. This trend is also visible in the contour plots of experimental data. However, the lines for the predicted density values intersect the corresponding lines of the experiments, thus the simulation provides an overall good agreement with the measured density distributions at all height levels in the compact.

The density measurement with the use of a  $\mu$ CT system and subsequent image processing possesses a few, but not substantial, limitations. Data from  $\mu$ CT imaging shows trends of an increase in density in the radial direction with a lower density in the center of the compact and a higher density at the outside. As shown in Fig. 6(a), the  $\mu$ CT images show that individual grains with an apparent higher density at the outer rim compared than at the center of the compact. However, grains irrespective of their position have the same density as they are made of the same material. We employed a threshold to reduce this effect. However, since the correlated images exhibit significant noise, changing the threshold also changes shape and size of the grains, and some of these artifacts appear to bias the density measurement. The quality of computer tomography depends on several factors, Stock [29]. The microstructure to be scanned should have a sufficient contrast in X-ray absorption. If constituents of similar atomic weight are scanned, the correlation computation might not be able to differentiate between them. Additionally, the spatial distribution of the constituents is a major influence. Every directional X-ray scan includes noise which the correlation computation has to reduce or eliminate to produce the absorption map. If the wavelength of the noise is on the same order as the wavelength of the microstructure, then the final image becomes blurred. This problem becomes especially dominant in scans of specimen with a small and evenly distributed void volume fraction, and additional uncertainty is introduced into the density measurements.

#### 4. Conclusion

An experimental method and related analytical equations for the determination of the material properties for powder compaction were proposed. Experiments were carried out for NaCl and the material properties characterizing NaCl powders with the Drucker–Prager–Cap model were determined. Such data can be useful as NaCl is often employed as a model material in designing and evaluating compaction processes. The material properties were then used in finite element simulations with the Drucker–Prager–Cap model. In parallel, NaCl

compacts were scanned using  $\mu$ CT and image processing, such that the spatial density distribution was determined experimentally. The results of the simulation were compared to the findings of traditional compaction experiments, e.g. load–displacement curves, and to those obtained by  $\mu$ CT and image processing.

Employing the material properties obtained by the present approach, simulations showed very good agreement to the experimental results for global characteristics. In particular two important quantities, the compaction stress–displacement relation and the residual strains are well predicted. Residual stresses due to these residual strains could significantly affect the performance of any component produced from a powder compaction process. Some details of the experiments could not be captured by the present model approach. Experiments show a sudden increase in load during the final stages of the loading. It appears that this is an effect of the discrete fracture of individual grains and the internal rearrangement of particles. A continuum simulation cannot capture such events. As such, model predictions above the critical compaction load are less accurately predicted.

Even on a local level, simulations and experiments are in good agreement. The relative increase in density obtained by simulations and the  $\mu$ CT together with image processing agree well. Potential alignment problems in the  $\mu$ CT apparatus and the overemphasis of the outside region in the  $\mu$ CT images can lead to deviations of the actual density and that measured from the proposed  $\mu$ CT analysis process.

In summary, by employing the Drucker–Prager–Cap model together with the novel approach for parameter identification, numerical simulation of powder compaction is able to provide significant support for manufacturing processes of granular materials.  $\mu$ CT and image processing provides a powerful tool for the analysis of spatial density distributions in powder compacts.

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#### References

- [1] I.S. Sandler, F.L. DiMaggio, G.Y. Baladi, Generalized cap model for geological materials, *Journal of Geotechnical Engineering* (1976) 683–699.
- [2] I. Aydin, B.J. Briscoe, K.Y. Sanliturk, The internal form of compacted ceramic components: a comparison of a finite element modeling with experiment, *Powder Technology* 89 (1996) 239–254.
- [3] D. Falgon, E. Vidal-Salle, J.C. Boyer, R. Peczkalski, J. Andrieu, Identification procedure of a hardening law for powder compaction, *Powder Technology* 157 (2005) 183–190.
- [4] A. Michrafy, D. Ringenbacher, P. Tchoreloff, Modeling the compaction behavior of powders: application to pharmaceutical powders, *Powder Technology* 127 (2002) 257–266.
- [5] J.C. Cunningham, I.C. Sinka, A. Zavaliangos, Analysis of tablet compaction. I. Characterization of mechanical behavior of powder and powder/tooling friction, *Journal of Pharmaceutical Sciences* 93 (2004) 2022–2039.
- [6] C.Y. Wu, O.M. Ruddy, A.C. Bentham, B.C. Hancock, S.M. Best, J.A. Elliott, Modeling the mechanical behavior of pharmaceutical powders during compaction, *Powder Technology* 152 (2005) 107–117.
- [7] C.Y. Wu, B.C. Hancock, A. Mills, A.C. Bentham, S.M. Best, J.A. Elliott, Numerical and experimental investigation of capping mechanisms during pharmaceutical tablet compaction, *Powder Technology* 181 (2008) 121–129.
- [8] G. Frenning, Analysis of pharmaceutical powder compaction using multiplicative hyperelasto-plastic theory, *Powder Technology* 172 (2007) 102–112.
- [9] M.H. Es-Saheb, Uniaxial strain rate effects in pharmaceutical powders during cold compaction, *Journal of Material Science* 27 (1992) 4151–4159.
- [10] M.S. Nielsen, N. Bay, M. Eriksen, J.I. Bech, B.C. Hancock, An alternative to the conventional triaxial compression test, *Powder Technology* 161 (2006) 220–226.
- [11] L.C.R. Schneider, A.C.F. Cocks, Experimental investigation of yield behavior of metal powder compacts, *Powder Metallurgy* 45 (2002) 237–245.
- [12] D.H. Phillips, J.J. Lannutti, X-ray computed tomography for testing and evaluation of ceramic processes, *American Ceramic Society Bulletin* 72 (1993) 69–75.
- [13] D. Bonnenfant, F. Mazerolle, P. Suquet, Compaction of powders containing hard inclusions: experiments and micromechanical modeling, *Mechanics of Materials* 29 (1998) 93–109.

- [14] C.M. Kong, J.J. Lannutti, Localized densification during the compaction of alumina granules, the stage I–II transition, *Journal of the American Ceramic Society* 83 (2000) 685–690.
- [15] I.C. Sinka, S.F. Burch, J.H. Tweed, J.C. Cunningham, Measurements of density variations in tablets using X-ray computed tomography, *International Journal of Pharmaceutics* 271 (2004) 215–224.
- [16] X. Fu, J.A. Elliott, A.C. Bentham, B.C. Hancock, R.E. Cameron, Application of X-ray micro-tomography and image processing to the investigation of a compacted granular system, *Particle and Particle Systems Characterization* 23 (2006) 229–236.
- [17] A. Marmottant, L. Salvo, C.L. Martin, A. Mortensen, Coordination measurement in compacted NaCl irregular powders using X-ray microtomography, *Journal of the European Ceramic Society* 28 (2008) 2441–2449.
- [18] J.S. Hardman, B.A. Lilley, Mechanisms of compaction of powdered materials, *Proceedings of the Royal Society of London A* 333 (1973) 183–199.
- [19] N.G. Lordi, H. Cocolas, H. Yamasaki, Analytical interpretation of powder compaction during loading phase, *Powder Technology* 90 (1997) 173–178.
- [20] J.M. Sonnergaard, A critical evaluation of the Heckel equation, *International Journal of Pharmaceutics* 193 (1999) 63–71.
- [21] O. Simon, P. Guigon, Correlation between powder-packing properties and roll-press compaction heterogeneity, *Powder Technology* 130 (2003) 257–264.
- [22] K.M. Picker, The 3D model: explaining densification and deformation mechanisms by using 3D parameter plots, *Drug Development and Industrial Pharmacy* 30 (2004) 413–425.
- [23] G. Alderborn, A novel approach to derive a compression parameter indicating effective particle deformability, *Pharmaceutical Development and Technology* 8 (2003) 367–377.
- [24] E. Doelker, D. Massuelle, Benefits of die-wall instrumentation for research and development in tableting, *European Journal of Pharmaceutics and Biopharmaceutics* 58 (2004) 427–444.
- [25] R.W. Davidge, Densification of NaCl powder in air, *Proceedings of the British Ceramic Society* 12 (1969) 245–249.
- [26] <http://www.saltinstitute.org/15.html>.
- [27] T. Kraft, H. Riedel, P. Stingl, F. Wittig, Numerische Simulation des Pressens und Sinterns, in: *Technische Keramische Werkstoffe*, J. Kriegesmann, ed., Deutsche Keramische Gesellschaft e.V., Fachverlag Deutscher Wirtschaftsdienst, Köln, Germany (1999) 1–17.
- [28] D.T. Gethin, R.W. Lewis, R.S. Ransing, A discrete deformable element approach or the compaction of powder systems, *Modeling Simulation in Material Science and Engineering* 11 (2003) 101–114.
- [29] S.R. Stock, Recent advances in X-ray microtomography applied to materials, *International Materials Reviews* 53 (2008) 129–181.